

Phase Noise in Beamforming

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Abstract—In this article we examine the effect of phase noise on a multi-antenna beamforming system. The effect is quantified in terms of error vector magnitude (EVM). It is shown that using independent phase noise sources improves EVM performance in the main lobe. It is further shown that when forming beams using antenna amplitude attenuation (tapering) for groups of correlated noise sources, lowest EVM is obtained when correlated noise sources are assigned to tapered amplitudes in a way which minimizes the differences between the sums of the tapering coefficients of the groups. Bounds for the variance of the phase noise in the main lobe are provided. Further a method achieving secure communication by forming a secure beam by adding uncorrelated phase noise to the antenna elements is presented.

Index Terms—Beamforming, phase noise, error vector magnitude, active antenna systems.

I. INTRODUCTION

PHASE noise originating from the analog components such as local oscillators (LO) of a multi-antenna system can be a major contributor to the performance limitations of that system. It can be so in terms of modifying the beam shape properties of a directed beam, or by degrading the effective communication channel and thereby the capacity of the system. For the first of those two views, [1] provides an extensive treatment on the influence of phase noise on beam sidelobe levels, main lobe shift and other parameters. In [2] the EVM is calculated for modulated signals distorted by phase noise and the effect on symbol error rate (SER) is analyzed, for systems without error coding or advanced receiver algorithms. The system capacity view then can be found in [3]: There the system relies on an architecture where one transmit-receive RF chain is switched into an antenna array for measurement purposes. Short-term phase noise in MIMO systems is modeled as an auto-regressive process, and verified in measurements. The impact on capacity is calculated and it is found that the capacity is dependent on the largest eigenvalue of the noise covariance matrix. Similarly [4] shows how phase noise in a measurement system affects the rank of the channel matrix and hence the estimated capacity. The authors in [5] estimate the phase noise of a MIMO system by means of pilots, and reduce the phase noise's impact by Wiener filtering. While separate or equal phase noise sources for the transceivers are considered, the effect of applying specific constant phase weights to the MIMO system (as in directed beamforming) is out of their scope.

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In a somewhat different focus in this paper we neither examine the impact of phase noise on beamshapes, nor possible receiver performance in terms of SER or bit error rate (BER) of a communication system, but how phase noise is transformed in the beamforming process and will affect the error vector magnitude (EVM) as seen at the receiver antenna, before any channel estimation or decoding processes.

Another aspect of this paper regards antenna arrays which employ antenna taper. Antenna taper is a technique where the excitation of the array is modified to reduce sidelobes for a directed beam [1], [6]. The implementation of such taper can be done by the physical layout of the antennas [7], or by setting coefficients in the signal processing unit of an active array. Our work leads to design recommendations for tapered antenna arrays, where phase noise sources such as local oscillators may be shared among antenna branches.

An introduction to the problem in mathematical terms is provided in Section II. In Section III we continue on the mathematical track to investigate the error vector caused by phase noise at the point of constructive interference. Simulations help to complement the analysis and illustrate the system behavior in Section IV. Section V presents an application of phase noise to achieve secure communications. Section VI concludes this paper.

II. PHASE NOISE IN BEAMFORMING

A. Superposition of Signals

Let us consider a beamforming system with transceivers, where the antenna elements are relatively close to each other. The received signal at a given time instance is a superposition of the signals of all antennas

$$s = \sum_{k=1}^n a_k e^{j(\Theta_k + \varphi_k)}, \quad (1)$$

where Θ_k comprises the phase shifts due to a carrier frequency or pre-set phase shift, a_k is the amplitude and φ_k is the phase noise. We assume that the phase noise changes so slowly with time that the flight time differences between the antenna elements can be neglected, that is the phase noise for all antenna elements is taken at the same time. The time series of phase noise values may be described as a realizations of a Gaussian random process of zero mean [4], [2], and we will regard φ_k as a Gaussian random variable associated with antenna k . Then (1) describes the effect of phase noise on a signal transmitted by a number of antennas as seen at one receive antenna, or a signal received by a number of antennas as seen after beamforming processing. Thus, we will examine the effect of phase noise of a transmit array, or that of a receive array, but not that of both at the same time.

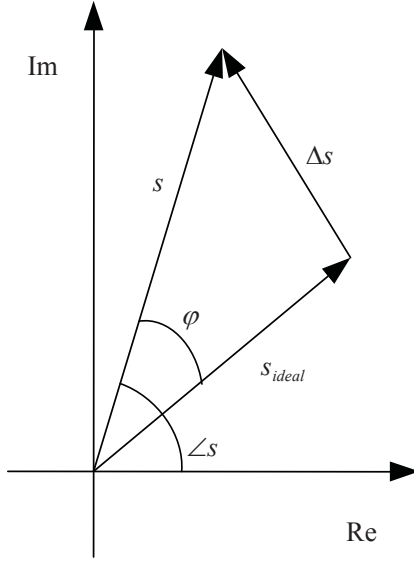


Fig. 1. Definition of the error vector Δs . s_{ideal} is the original complex vector distorted by phase noise φ and some amplitude error resulting in the vector s .

When the phase noise is fully correlated in the different transmitters, that is when $\varphi_k = \varphi \forall k$ the signal is

$$s = \sum_{k=1}^n a_k e^{j(\Theta_k + \varphi)} = e^{j\varphi} \sum_{k=1}^n a_k e^{j\Theta_k} = e^{j\varphi} s_{ideal}, \quad (2)$$

where the exponent rotates the complex signal vector by angle φ while keeping the amplitude constant. Thus in the case of correlated phase noise the effect is similar as in a one-antenna system, that is the signal amplitude is not affected and the phase noise simply rotates the received signal phase.

B. Error Vector Magnitude

Error vector magnitude (EVM) is a quantity frequently used to characterize the effect of phase noise on the linearity of a transceiver and/or on the constellation points of transmitted complex symbols. It may be defined in various ways, here we will use the receiver architecture ignorant definition of the error vector Δs [2].

The error vector Δs is obtained by subtracting the ideal signal from (1) as in Fig. 1,

$$\begin{aligned} \Delta s &= s - s_{ideal} = \sum_{k=1}^n a_k e^{j(\Theta_k + \varphi)} - \sum_{k=1}^n a_k e^{j\Theta_k} \\ &= \sum_{k=1}^n a_k e^{j\Theta_k} (e^{j\varphi} - 1). \end{aligned} \quad (3)$$

We can calculate its root mean square value as

$$EVM_{RMS}^2 = \frac{E[|\Delta s|^2]}{|s_{ideal}|^2} = \frac{\text{var}(\Delta s) + |E[\Delta s]|^2}{|s_{ideal}|^2}, \quad (4)$$

where var denotes variance and $E[\cdot]$ the mean value. The RMS value here is scaled with the length of the ideal signal vector, as it is often expressed in %.

When the phase noise is fully correlated, the error vector of (3) simplifies to

$$\Delta s_{corr} = (e^{j\varphi} - 1) \sum_{k=1}^n a_k e^{j\Theta_k} = (e^{j\varphi} - 1) s_{ideal}, \quad (5)$$

and the RMS value is given by

$$\begin{aligned} EVM_{RMS}^2 &= \frac{|s_{ideal}|^2 E[|(e^{j\varphi} - 1)|^2]}{|s_{ideal}|^2} = 2E[1 - \cos\varphi] \\ &\approx E[\varphi^2] = \sigma_\varphi^2. \end{aligned} \quad (6)$$

In (6) the EVM is independent of both the phases Θ_k and the number of antennas n . With small values of phase noise φ we can take the Taylor expansion and then the EVM reduces to the standard deviation σ_φ of the phase noise. In situations where it is assumed that the major contributor to the EVM is phase noise one may see the EVM therefore scaled to degrees.

The error caused by using the Taylor expansion increases with the square of the phase noise being about 5% with phase noise around 45 deg, and about 0.3% with a 10 deg phase noise.

III. POINT OF CONSTRUCTIVE INTERFERENCE IN DIRECTED BEAMFORMING

A. Error Vector Magnitude at the Point of Constructive Interference

Considering only the signal in the main lobe, that is when the phases of all antennas align in a point of constructive interference, that is when $\Theta_k = 0 \forall k$, it is possible to simplify (1) and obtain more concrete results for the EVM. For the case with uncorrelated phase noise the error vector magnitude is (see Appendix A)

$$EVM_{uncorr,RMS}^2 = \sigma_x^2 \frac{\sigma_a^2}{n\bar{a}^2} + \frac{\sigma_x^2}{n} + |\mu_x|^2, \quad (7)$$

and for the case with correlated phase noise

$$EVM_{corr,RMS}^2 = \sigma_x^2 + |\mu_x|^2, \quad (8)$$

where we introduced the abbreviation $x_k = (e^{j\varphi_k} - 1)$, its standard deviation $\sigma_x = \sigma_{x_k}$, and mean $\mu_x = \mu_{x_k}$. With reasonable phase noise values μ_x^2 is much smaller than σ_x^2 being about less than 1% when phase noise is 10 deg and less than 10% when phase noise is 35 deg. In addition we have denoted with $\bar{a}$ the average of the amplitude coefficients and their standard deviation with σ_a .

Using the Cauchy-Schwartz inequality and assuming $\mu_x \approx 0$ it can be shown (see Appendix A) that for the main lobe the EVM for uncorrelated noise sources relates to that of correlated ones as

$$EVM_{uncorr,RMS}^2 \leq EVM_{corr,RMS}^2 \leq n EVM_{uncorr,RMS}^2. \quad (9)$$

The largest ratio between correlated and uncorrelated sources is obtained for the special case where all the amplitudes are the same, that is when $a_k = a \forall k$. That is also the condition under which the EVM is minimal.

As can be seen from (8) the EVM of the correlated case does not depend on the values of the amplitudes, while (7)

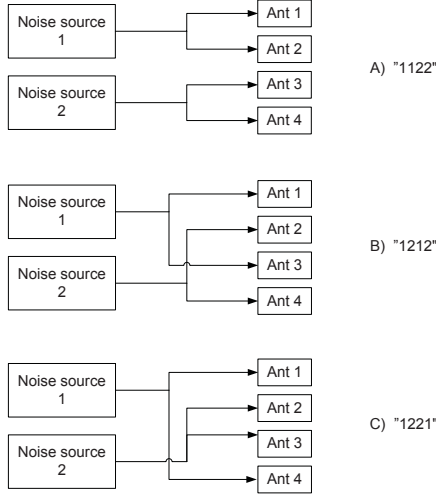


Fig. 2. Tree ways of coupling two noise sources to four antennas.

shows that for the uncorrelated they do. If we consider the case where the sum of amplitudes is constant, the EVM is minimized when the variance of a_k is zero, that is when the amplitudes are all equal. This indicates that the EVM cannot be improved by distributing amplitudes in any other fashion. If we consider the case where the sum of amplitudes squared is constant, that is the constant power case, then the minimum EVM is obtained when $\bar{a}^2 = \frac{1}{n} \sum_{k=1}^n a_k^2 - \sigma_a^2$ is at maximum, which again happens when the variance of the amplitudes is 0.

While it may be somewhat unintuitive that the EVM decreases with the use of independent noise source the physical explanation is that at the point of constructive interference uncorrelated phase noise averages out, and the signal vector will divert less from its ideal direction than in the case where the signal components of all antennas have the same phase error.

B. Error Vector Magnitude With Groups of Correlated Phase Noise

In some transceiver designs there will be more than one LO, and one LO may serve more than one antenna branch, but not all. The effect is that there will be groups of antenna branches which have correlated phase noise. Also, there will be multiple ways to make the connections between the antennas of the array, and the group of oscillators. In Fig. 2, we show all the possible connections for an example of four antennas and two noise sources each feeding two antennas. In a system with n antennas and m phase noise sources there are

$$N = \frac{n!}{((n/m)!)^m m!} \quad (10)$$

different ways to connect the noise sources to the array. The number of combinations increases rapidly when more antennas are added to the system. In Appendix B we show that for groups of correlated phase noise sources the EVM is

$$EVM_{comb,RMS}^2 = \sigma_x^2 \frac{\sigma_a^2}{m \bar{a}^2} + \frac{\sigma_x^2}{m} + |\mu_x|^2, \quad (11)$$

which is identical in shape to (7). Analogous to Section III-A the EVM is minimized when the groups of correlated phase noise sources are assigned to the antennas in such a way that the sum of the antenna tapering coefficients of each group is equal to that of any other group. As an example let us consider the special situation of four antennas and two noise sources (see Fig. 2). Let us assume further that the amplitude coefficients are symmetric, that is $a_1 = a_4 \wedge a_2 = a_3$ and that $\mu_x \approx 0$. Then using (11) we get for the EVM in cases A) and B)

$$EVM_{AB,RMS}^2 = \frac{\sigma_x^2}{2}, \quad (12)$$

which is independent of the antenna coefficients a_k , and for the case C)

$$EVM_{C,RMS}^2 = \frac{a_1^2 + a_2^2}{a_1^2 + 2a_1a_2 + a_2^2} \sigma_x^2. \quad (13)$$

The EVM of case C) is always larger than or equal to the cases A) and B), because the term $\frac{a_1^2 + a_2^2}{a_1^2 + 2a_1a_2 + a_2^2}$ preceding σ_x^2 in (13) takes on values between 0.5 (when $a_1 = a_2$) and 1 (when one of the coefficients is zero).

Another view on the same result in that context is that the amount of reduction in phase noise energy that can be achieved at the point of constructive interference will depend on the amount of tapering present in the system, and can be computed from the tapering amplitudes according to (11). The gains will be quantified for an example in the following section.

Further, since we do not make any assumptions on the geometric arrangements of the antennas this result holds also for 2- or 3-dimensional antenna arrays. In a case where the antenna amplitudes are fixed and we are looking for the optimal way to connect $m = N/2$ phase noise sources to N antennas the lowest error vector variance is obtained when the pairs are successively assigned to highest and lowest tapering coefficients (Appendix C).

IV. ERROR VECTOR MAGNITUDE IN SIMULATIONS

A. Simulation Setup

In the previous sections we examined some of the behavior of the beamformed noise in mathematical terms. In the following we show simulation results and visualize error vector behavior for directed beams. In the mathematical analysis we concentrated on the main lobe / point of constructive interference, and the simulation will allow us to verify those results. Furthermore, with simulations we are able to characterize the EVM behavior for not only one point, but for all observation angles, even if only for the selected simulated example. The configuration chosen for simulation was a system with four antennas where phase noise was added to the antennas according to the phase noise grouping scheme under study. The array was assumed to be a linear antenna array with antenna elements spaced apart by $\lambda/2$. The beamformed signal s_θ is given by

$$\begin{aligned} s_\theta &= \cos \theta \sum_{k=1}^4 a_k e^{j(\Theta_k - \pi k \sin \theta + \varphi_k)} \\ &= \cos \theta \sum_{k=1}^4 a_k e^{j(\pi k \sin \Gamma - \pi k \sin \theta + \varphi_k)}, \end{aligned} \quad (14)$$

TABLE I

NORMALIZED EVM WITH $\Phi_{RMS,org} \approx 4$ DEG, FOR A DIRECTED BEAM AT THE POINT OF CONSTRUCTIVE INTERFERENCE, FOR DIFFERENT NOISE ORDERINGS, SIMULATED AND THEORETICAL VALUES. THE NORMALIZATION HERE SERVES THE PURPOSE OF FACILITATING THE COMPARISON, BUT ALSO INDICATES THE PERCENTAGE OF EVM REDUCTION WITH RESPECT TO FULLY CORRELATED NOISE SOURCES.

	EVM_{RMS}/σ_x at $\theta = \Gamma$, simulated	EVM_{RMS}/σ_x at $\theta = \Gamma$, using (11)
"1 2 3 4"	0.528 ± 0.001	$\sqrt{5/18} = 0.527$
"1 2 2 1"	0.745 ± 0.001	$\sqrt{5/9} = 0.745$
"1 1 2 2"	0.705 ± 0.001	$\sqrt{1/2} = 0.707$
"1 2 1 2"	0.707 ± 0.001	$\sqrt{1/2} = 0.707$

where θ is the azimuth angle of observation, Γ is the angle to which the beam is directed and k the antenna index. Antenna elements had a $\cos\theta$ radiation pattern [8]. The phase noise spectrum had an energy (meaning also phase noise variance) of $\Phi_{RMS,org} \approx 4$ deg. For generating the noise a complex AWGN signal was generated, shaped with the power spectral density (PSD) mask, and transferred to the time domain, [5], [9]. The PSD mask was chosen to reflect a typical shape of a low-pass filter with cut-off frequency of 20kHz [10], [11], [12]. For most of the simulations the beam direction was $\Gamma = 40$ deg and we used a symmetric amplitude vector $a_1 \dots a_4 = [0.5 \ 1.0 \ 1.0 \ 0.5]$.

Note that in Section II we derived the relation between EVM and phase noise energy for the assumption of white noise. When shaping the spectrum of the noise a time dependency is introduced, and part of the derivation may not be applicable any longer, e.g. when the random variables are added. However we observed little effect of the shape of the PSD on the results and therefore chose to include the shaping into simulations, so as to stay closer to real world phase noise spectra.

B. Phase and Amplitude Histogram, and Error Vector Magnitude

In the following we show simulation results of noise constellations for systems presented in Fig. 2. For comparison also the case of all antenna elements having uncorrelated noise is evaluated. Fig. 3 shows the color coded histogram for the example of the mirror symmetric "1221" constellation. The solid lines in the power plots indicate the ideal case (lower line) and the RMS power of the noisy signal (upper line). We observe that for this ordering "1221", the beamshape exhibits a vanishing minimum. In Fig. 4 we show the error vector magnitudes from the same simulations, including the case with fully correlated phase noise. The figure shows that phase and amplitude variations are small in the main beam region. The behavior in sidelobe regions depends heavily on the correlations of the phase noise sources.

In Table I we compare the EVM values obtained from simulations and equation (11), where we have assumed that $\mu_x \approx 0$ (the error caused by that approximation is about 1%). The plots of the EVM in Fig. 4 also show that for some constellations the EVM stays low over longer azimuthal range

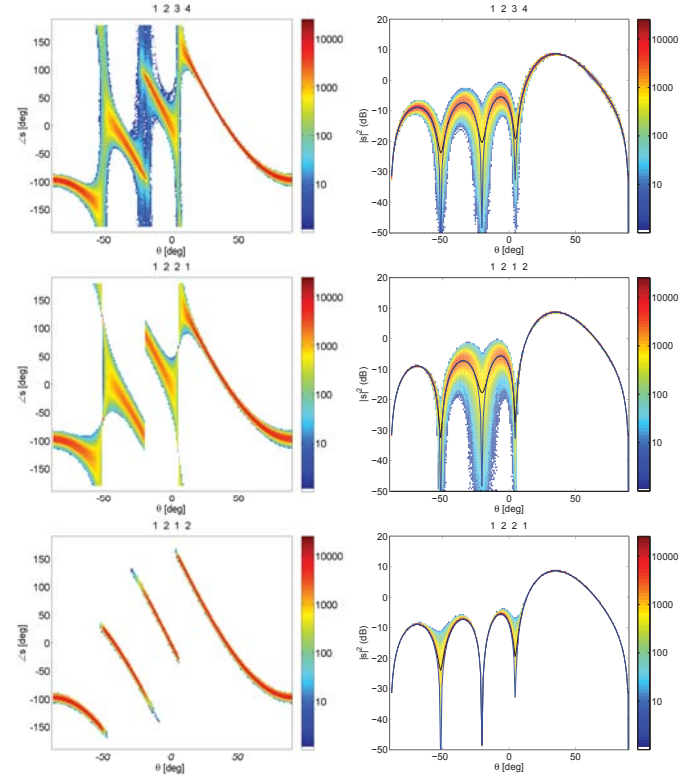


Fig. 3. Color coded histogram of beam phases and amplitudes, for a 4 antenna beam directed at 40 deg. The topmost plots show the usage of four independent noise sources, whereas in the lower ones there are two noise sources feeding each two antennas.

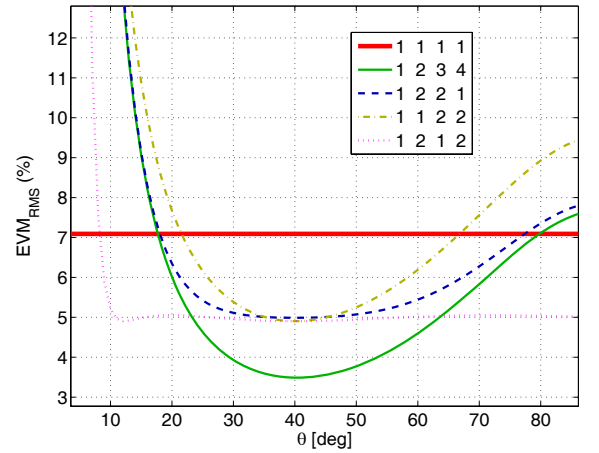


Fig. 4. EVM of a beam directed at 40 deg.

than for others, especially for the ordering "1212". Table II is therefore presenting values for the phase noise and EVM averaged over the half power beam width (HPBW). As shown before, the EVM will depend on tapering coefficients and noise power, and therefore the table Table II should be read only to indicate a trend. In Table II we are further introducing

$$\Phi_{RMS} = \sqrt{\text{var}(\mathcal{L}s - \mathcal{L}s_{ideal})} = \sqrt{\text{var}(\varphi)}, \quad (15)$$

which is the standard deviation of the phase error and can be also understood as the effective phase noise amplitude. $\Phi_{RMS,org}$ is the phase noise power present at a single antenna and corresponds to σ_x for small φ . Fig. 5 shows Φ_{RMS} for the

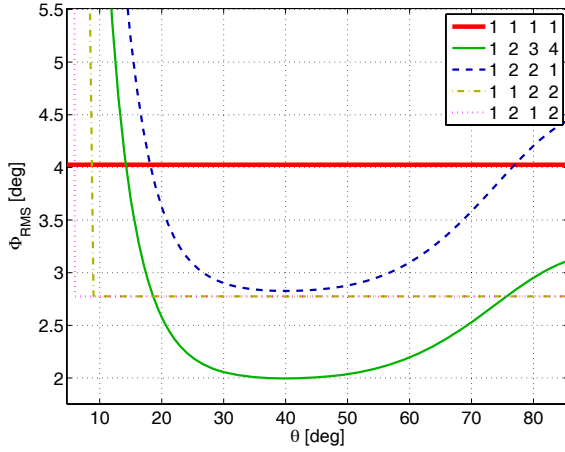


Fig. 5. Standard deviation of the phase error of the beamformed signal caused by the phase noise.

TABLE II

PHASE NOISE AND EVM WITH $\Phi_{RMS,org} \approx 4$ DEG, FOR A DIRECTED BEAM AT THE POINT OF CONSTRUCTIVE INTERFERENCE AND FOR THE HPBW, FOR DIFFERENT NOISE ORDERINGS. THE EVM VALUES ARE SCALED BY $180/\pi$ TO BE COMPARABLE WITH THE PHASE NOISE ENERGIES.

	Φ_{RMS} at $\theta =$ Γ [deg]	mean Φ_{RMS} (HPBW) [deg]	mean EVM (HPBW) [deg]
"1 2 3 4"	2.00 ± 0.01	2.11 ± 0.01	2.34 ± 0.01
"1 2 2 1"	2.83 ± 0.01	2.98 ± 0.01	3.01 ± 0.01
"1 1 2 2"	2.77 ± 0.01	2.77 ± 0.01	3.19 ± 0.01
"1 2 1 2"	2.77 ± 0.01	2.77 ± 0.01	2.84 ± 0.01

beamformed signal and thus provides a look at the resulting phase noise only, excluding effects of amplitude errors. We observe that the beamformed phase noise standard deviation is constant for a wide azimuthal range for the orderings "1122" and "1212". This comes about by the symmetric assignment of phase noise sources to antenna element vectors of equal magnitude. In Table II we also show the mean EVM and Φ_{RMS} calculated in the HPBW region. It can be seen that within the HPBW the major part of the EVM is caused by the phase error, not the amplitude error. Besides showing the good match between simulations and equations, we may here also use the tables to exemplify the gains that can be obtained for the optimal phase noise grouping: The case "1212" has a 5% better EVM performance than the case "1221" for the point of constructive interference (Table I) and a 10% gain for the mean EVM in the HPBW (Table II). Those gains can be used to relax phase noise requirements on oscillators in the system without degrading EVM performance compared to a system with "1221" ordering.

C. Phase Noise Power Spectral Density Change

We mentioned earlier in Section IV-A that the shape of the phase noise PSD does not have an effect on the EVM, but that the beamforming in principle may alter the phase noise PSD (even though the spectrum of the beamformed signal will not be affected). We therefore measured $\varphi = \mathcal{L}s - \mathcal{L}s_{ideal}$

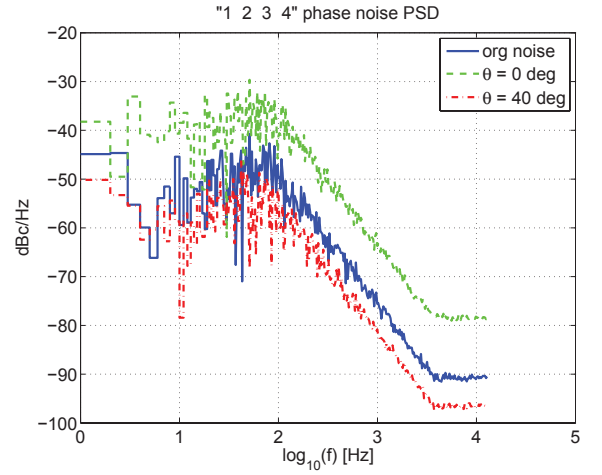


Fig. 6. Spectrum of phase noise, original before beamforming (blue curve) and after beamforming at two different angles (red and green curves).

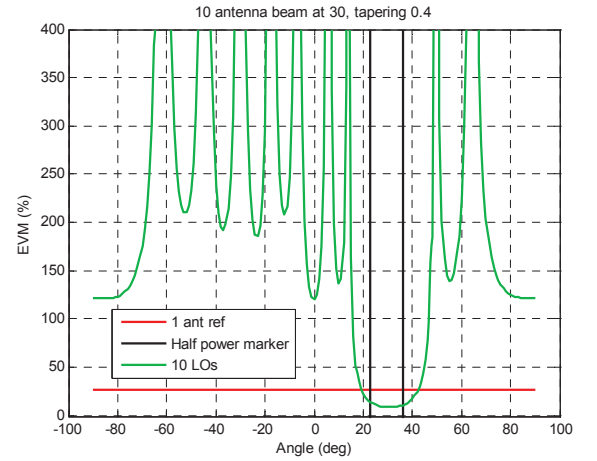


Fig. 7. Effect of heavy phase noise on 10 antenna directed beam.

in the simulation for two observations angles, the point of constructive interference at $\theta = 40$ deg, and a point of minimal beam power at $\theta = 0$ deg, and are displaying its FFT in Fig. 6. Note that this excludes the amplitude error that is caused by the original phase noise during the beamforming. The figure shows a significant increase around beam minimum, as can be expected since the beam amplitudes are very small and thus the phase can change easily. The change of noise power in the PSD corresponds to that shown in Fig. 5. The new observation here is that the phase noise PSD does not change its shape. This suggests that the nonlinear addition of phases is negligible.

V. SECURE BEAMFORMING

In the previous section we saw that the phase-noise generated error - amplitude and phase - is smallest in the main lobe. While typically one will want to combat the noise, in some applications it might be also desirable to add even more of it, for instance to achieve communication which is difficult to eavesdrop. The example in Fig. 7 show the effect of phase noise on the EVM in cases of directed beamforming with 10 antennas. The important thing to note here is that the EVM

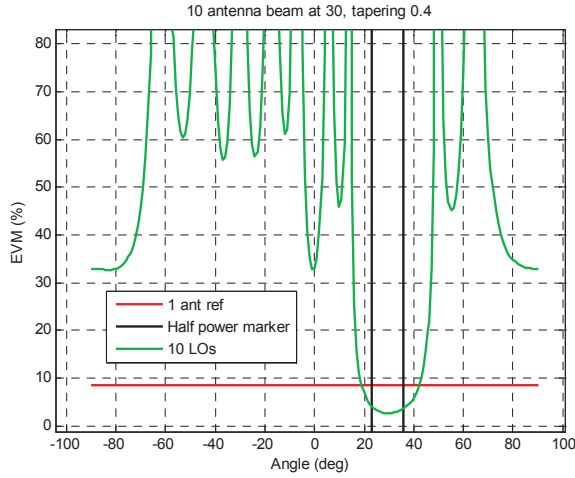


Fig. 8. Effect of phase noise on 10 antenna directed beam, same as Fig. 7 but with less phase noise.

is over ten times larger outside the main beam (here at 30 deg) than in the main beam. Also this relative difference in EVM remains when the amount of phase noise is changed, see Fig. 8.

The effect of phase noise is to limit how big the final signal to noise ratio (SNR) can be. Thus even if the signal is transmitted with high power and the level of other noise and interference that the receiver sees is very small the SNR cannot go above what is dictated by the added phase noise. The SNR relates to the EVM depending on the chosen receiver architecture. In fact, in many systems EVM is defined after receiver channel estimation, a step that involves phase noise removal. Nevertheless, there will be a point where the receiver is not able to cope with the phase noise any longer. The secure communication can be established by designing the system so that within the main lobe that point is just not yet reached. It is thus possible to make a beamforming system where the signal is easily decodable in the main lobe, but unreadable in the side lobe regions by having enough uncorrelated phase noise in the system. One way to do it is to use a separate LO for each transmission chain ensuring that the phase noise is at least partly independent in the different antenna elements. The phase noise could then be increased to desired level by using e.g. lower quality LOs. Another implementation would be to employ a dedicated phase noise source, possibly with a tuning option to change the magnitude of the noise. One disadvantage of the system is that the average power radiated to regions outside the main lobe is increased by the addition of phase noise. Further the added phase noise could be present only in portions of the signal, which need to be protected. Note that in many system pilots are used to track the phase error [13], [14], [15], and our above considerations do not take into account the pilot-aided phase tracking. However using pilots requires having achieved synchronization, something that may not be possible with above described system. Also, one may change the shape of phase noise spectrum to higher frequencies, making the task of tracking and channel estimation more difficult.

VI. CONCLUSION

In this article we examined the effect of phase noise on a beamforming system. The focus was to study phase noise originating from local oscillators of the transceivers of active antennas, and its effect on the error vector of the beamformed signal. In particular we were investigating directed beamforming where phases are aligned so as to produce constructive interference in a certain direction. We showed that the error vector is smallest in the direction of interest, if uncorrelated phase noise sources are used. When groups of correlated noise sources are present, the way they are connected to the antennas has an impact on error performance. Algorithms for mapping noise sources to antennas were derived which achieve minimal error vector. The implication is that in certain circumstances system design parameters can be relaxed. For instance, it is common practice to over-specify necessary sidelobe levels for antenna arrays, to account for phase noise [1], chapter 7.2. Or, when leaving beam parameters intact, one may relax the requirements on frequency synthesizers. The usage of independent noise sources should however be considered in any transceiver algorithms which have independent payload signals on different antennas. Future work could concentrate on examining the effect of phase noise when taking into account also different receiver architectures, or systems employing blind or pilot aided channel estimation techniques.

APPENDIX A

ERROR VECTOR AT THE POINT OF CONSTRUCTIVE INTERFERENCE

For all phases in (3) aligning the error vector becomes

$$\Delta s = \sum_{k=1}^n a_k (e^{j\varphi_k} - 1) = \sum_{k=1}^n a_k x_k \quad (16)$$

where we have simplified the notation by using

$$x_k = (e^{j\varphi_k} - 1). \quad (17)$$

In order to calculate the error vector magnitude we first calculate the variance of the error vector for both, uncorrelated, and correlated phase noise, resulting in

$$\begin{aligned} \text{var}(\Delta s_{\text{uncorr}}) &= \sum_{k=1}^n \text{var}(a_k x_k) = \sum_{k=1}^n a_k^2 \text{var}(x_k) \\ &= \sigma_x^2 \sum_{k=1}^n a_k^2, \end{aligned} \quad (18)$$

and

$$\begin{aligned} \text{var}(\Delta s_{\text{corr}}) &= \text{var}\left(x_0 \sum_{k=1}^n a_k\right) = \sigma_x^2 \left(\sum_{k=1}^n a_k\right)^2 \\ &= \sigma_x^2 s_{\text{ideal}}^2, \end{aligned} \quad (19)$$

where we have denoted the variance of x_k as σ_x^2 and the ideal signal is $s_{\text{ideal}} = \sum_{k=1}^n a_k$. Using (4), (16), and (18) we get

for the error vector magnitude

$$\begin{aligned} EVM_{uncorr,RMS}^2 &= \frac{\sigma_x^2 \sum_{k=1}^n a_k^2}{\left(\sum_{k=1}^n a_k\right)^2} + |\mu_x|^2 \\ &= \sigma_x^2 \frac{\sigma_a^2 + \bar{a}^2}{n\bar{a}^2} + |\mu_x|^2 \\ &= \sigma_x^2 \frac{\sigma_a^2}{n\bar{a}^2} + \frac{\sigma_x^2}{n} + |\mu_x|^2 \end{aligned} \quad (20)$$

and

$$EVM_{corr,RMS}^2 = \sigma_x^2 + |\mu_x|^2. \quad (21)$$

Note that from (5) we see that (21) applies also to cases with θ_k having any values. We can get an upper limit for the correlated EVM by using Cauchy-Schwartz inequality

$$\left(\sum_{k=1}^n a_k\right)^2 \leq \sum_{k=1}^n 1^2 \sum_{k=1}^n a_k^2 = n \sum_{k=1}^n a_k^2 \quad (22)$$

with the variance of the uncorrelated error vector (18)

$$\begin{aligned} \text{var}(\Delta s_{uncorr}) &= \sigma_x^2 \sum_{k=1}^n a_k^2 \\ &\geq \frac{\sigma_x^2}{n} \left(\sum_{k=1}^n a_k\right)^2 = \frac{\text{var}(\Delta s_{corr})}{n}. \end{aligned} \quad (23)$$

Further

$$\begin{aligned} \left(\sum_{k=1}^n a_k\right)^2 &= \sum_{k=1}^n \sum_{l=1}^n a_k a_l = \sum_{k=1}^n a_k^2 + \sum_{k=1}^n \sum_{l=1, l \neq k}^n a_k a_l \\ &\geq \sum_{k=1}^n a_k^2 \end{aligned} \quad (24)$$

and thus we see from (18) and (19) that the variance of the error vector is always smaller in the uncorrelated case. Combining these and assuming that the phase noise is relatively small, with (17) it happens that $\mu_x \approx 0$ and

$$EVM_{uncorr,RMS}^2 \leq EVM_{corr,RMS}^2 \leq n EVM_{uncorr,RMS}^2. \quad (25)$$

In the special case where all the amplitudes are the same, that is when $a_k = a \forall k$ and thus $\sigma_a = 0$, the error vector magnitude for the uncorrelated case reduces to

$$EVM_{uncorr,RMS}^2 = \frac{\sigma_x^2}{n} + |\mu_x|^2. \quad (26)$$

APPENDIX B

GROUPING WITH m PHASE NOISE SOURCES

Consider the case where groups of n/m antennas share the same m noise source. Using (16) we get for the error vector

$$\Delta s = \sum_{k=1}^m \left(x_k \sum_{l=1}^{n/m} a_{k,l} \right) = \frac{n}{m} \sum_{k=1}^m (x_k \bar{a}_k), \quad (27)$$

where the first sum is over the different uncorrelated noise sources and the second sum over the antennas connected to the noise source k . Since the phase noise in the antennas connected to a single noise source are correlated, x does not

depend on l and $\bar{a}_k$ is the average amplitude inside one noise source. From this we can calculate the variance of the error vector,

$$\begin{aligned} \text{var}(\Delta s) &= \sum_{k=1}^m \text{var} \left(x_k \frac{n}{m} \bar{a}_k \right) = \sum_{k=1}^m \left(\frac{n}{m} \bar{a}_k \right)^2 \text{var}(x_k) \\ &= \sigma_x^2 \frac{n^2}{m^2} \sum_{k=1}^m \bar{a}_k^2, \end{aligned} \quad (28)$$

where we have assumed that the variance of x_k is the same for all noise sources. The EVM is then obtained using (4) as

$$\begin{aligned} EVM_{comb,RMS}^2 &= \frac{\sigma_x^2 \frac{n^2}{m^2} \sum_{k=1}^m \bar{a}_k^2 + \left(|\mu_x| \frac{n}{m} \sum_{k=1}^m \bar{a}_k \right)^2}{\left(\frac{n}{m} \sum_{k=1}^m \bar{a}_k \right)^2} \\ &= \sigma_x^2 \frac{\sum_{k=1}^m \bar{a}_k^2}{\left(\sum_{k=1}^m \bar{a}_k \right)^2} + |\mu_x|^2 \\ &= \sigma_x^2 \frac{\sigma_{\bar{a}}^2}{m\bar{a}^2} + \frac{\sigma_x^2}{m} + |\mu_x|^2 \end{aligned} \quad (29)$$

which is identical in shape to (20).

APPENDIX C

GROUPING WITH $N/2$ NOISE SOURCES

Consider the case where there are N antennas and $m = N/2$ noise sources. Then the error variance (28) is

$$\text{var}(\Delta s) = \sigma_x^2 4 \sum_{k=1}^m \bar{a}_k^2, \quad (30)$$

and the $\bar{a}_k$ are pairs of coefficients associated with independent noise sources.

Conjecture: Lowest error variance is obtained when choosing for the first pair of $\bar{a}_k$ the lowest and highest values among a_n , for the second pair again the lowest and the highest a_n of the remaining set, and so on. That is, forming $\bar{a}_k$ as

$$\begin{aligned} \bar{a}_1 &= (a_1 + a_N) \\ \bar{a}_2 &= (a_2 + a_{N-1}) \\ &\vdots \\ \bar{a}_m &= (a_m + a_{m+1}) \end{aligned} \quad (31)$$

should lead to minimal

$$\begin{aligned} \text{var}(\Delta s) &= \sigma_x^2 4 \sum_{k=1}^{N/2} \bar{a}_k^2 \\ &= \sigma_x^2 4 \left(\sum_{k=1}^N a_k^2 + 2 \sum_{k=1}^{N/2} a_k a_{N-k+1} \right) \\ &= \sigma_x^2 4 \left(\sum_{k=1}^N a_k^2 + \sum_{k=1}^N a_k a_{N-k+1} \right) \end{aligned} \quad (32)$$

when $\{a_1, \dots, a_N\}$ are sorted in ascending order.

The term $\sum_{k=1}^N a_k^2$ will appear in for any way of sorting $\{a_1, \dots, a_N\}$. Therefore it remains to show that

$$\sum_{k=1}^N a_k a_{N-k+1} \quad (33)$$

is minimal for sorted $\{a_1, \dots, a_N\}$ only. Now recall that the rearrangement inequality [16] states that for a set of sorted real numbers u_k and v_k

$$u_N v_1 + \dots + u_1 v_N \leq u_{p(1)} v_1 + \dots + u_{p(N)} v_N \quad (34)$$

for every permutation $u_{p(1)}, \dots, u_{p(N)}$. That is, it states that $\sum_{k=1}^N u_k v_{N-k+1}$ is minimal for all ways of pairing u_k and v_k . With the special case of $u_k = v_k$ this is exactly what we have in (33).

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